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Software Development Group Project 5COSC021C

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# CW1 Group Project - PhysioConnect

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# I. Declaration

We hereby certify that this project and all the content associated with it is our own work and it has not been submitted before nor is currently being submitted for any other degree programme.

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# II. Abstract + Keywords

This report presents the development of a physiotherapy application designed to improve the physical health and recovery of users. Chapter 1 provides an introduction to the project, detailing the purpose, objectives, and the target audience of the application. It outlines the key features of the app, which include video consultations, personalized exercise plans, pain monitoring, and other functionalities aimed at enhancing the rehabilitation process. The chapter also introduces the stakeholders and explains the relevance of the application within the context of modern healthcare solutions.

Chapter 2 offers a comprehensive review of existing work in the field of physiotherapy applications. Through benchmarking and an analysis of current solutions, this chapter identifies the strengths and weaknesses of existing apps. It further discusses how the proposed solution can address gaps in the market and deliver superior user experience and functionality, positioning the app as a valuable tool in the physiotherapy space.

Chapter 3 details the methodology employed throughout the project. It outlines the research methods and approaches used to gather user requirements and design the system. The chapter also explains the development process, including the selection of appropriate technologies, design principles, and tools to ensure the success of the application. Emphasis is placed on the strategies used for system implementation, testing, and continuous improvement to deliver a user-centric, reliable solution.

**Keywords**: Physiotherapy app, healthcare solutions, benchmarking, user requirements, system design, application development.

# III. Acknowledgement

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# VII. Abbreviations table

SRS: System Requirements Specification

UI/UX: User Interface/User Experience

AR: Augmented Reality

AI: Artificial Intelligence

EPHR: Electronic Personal Health Record

API: Application Programming Interface

DBMS: Database Management System

DFD: Data Flow Diagram

MVC: Model-View-Controller

CRUD: Create, Read, Update, Delete

BCS: British Computer Society

KPI: Key Performance Indicator

HTTPS: Hypertext Transfer Protocol Secure

QA: Quality Assurance

GDPR: General Data Protection Regulation

MVP: Minimum Viable Product

CPU: Central Processing Unit

RAM: Random Access Memory

SQL: Structured Query Language

CI/CD: Continuous Integration/Continuous Deployment

# Chapter 1: Introduction

## 1.1 Chapter Overview

With a focus on concerns including high costs, time consumption, and a lack of knowledge about home-based therapy possibilities, this chapter explores the many obstacles that stand in the way of accessible and efficient DIY methodologies. Regular treatment sessions are unsustainable for many patients, including sportsman, everyday workers experiencing posture-related pain, the elderly having age-related issues, daily laborers recovering from physical strains and many more individuals due to the high session expenses, these can range up to LKR 3,500 to LKR 8,500 approximately in Sri Lanka or as much as $150 in the US. Consistency and recovery speed are eventually impacted by the additional strain that frequently commutes for sessions, which limits mobility, higher pain, inability to physically engage in work, etc.

## 1.2 Problem background

### 1.2.1 The Problem

The need for consistent and accessible physiotherapy related resources to support athletes in managing their injuries and tracking their recovery is critical at any venue and time. There are reportedly over 8 million sport related injuries reported in the US alone (National safety council, 2023). The major problem lies in trying to receive timely and consistent care. While professional physiotherapy services are invaluable, they are often limited by geographic location, cost, and time constraints, making it difficult for athletes to maintain a consistent recovery routine. Furthermore, in certain situations athletes are limited to a time span of a few minutes to hours (For example, getting a leg sprain a few hours before an important race). This is especially true for amateur athletes who may not have regular access to physiotherapists. The need for a solution that empowers athletes to manage their own recovery is clear.

This gap in accessible, reliable recovery guidance creates frustration and confusion for athletes

who are eager to recover quickly and return to their sports. Moreover, the lack of proper self-care education leaves athletes at risk of re-injury, as they may not know the best exercises to promote healing or understand the importance of injury-specific treatments, such as kinesiology taping or stretching routines. The widespread nature of these challenges calls for a solution that bridges this accessibility gap and provides athletes with the tools they need to manage their recovery safely and effectively.

1.2.2 Example Scenarios

**Scenario 1: Professional Athletes with High Physical Demands**

Athletes push their bodies to the limit, which can lead to injuries and persistent pain if not carefully managed. For example, a basketball player who trains daily may experience chronic knee pain from repetitive jumping, while a tennis player could struggle with elbow inflammation from repeated swings. These athletes often lack time for full rehabilitation due to demanding schedules, making a self-care app invaluable. The app could offer targeted tutorials on muscle recovery and tailored exercises to strengthen vulnerable areas, allowing athletes to address their discomfort independently. By monitoring pain levels and tracking their recovery progress, athletes gain greater control over their health, helping them reduce injury recurrence and prolong their careers.

**Scenario 2: Software Developers with Sedentary Work Conditions**

Software developers often spend hours at a computer, leading to issues like neck pain, backaches, and wrist strain from poor posture and repetitive movements. For instance, a developer working on a complex project might find themselves sitting for eight or more hours with minimal breaks. This lifestyle can create muscular imbalances and stiffness, which, if left unmanaged, may escalate into more severe conditions. A physiotherapy app could help by suggesting regular stretching routines, posture corrections, and exercises to improve circulation. Real-time reminders to take movement breaks and ergonomic tips could be especially helpful, enabling developers to manage discomfort and maintain productivity without requiring extensive in-person physiotherapy sessions.

**Scenario 3: Retirees Managing Age-Related Musculoskeletal Issues**

Retired individuals often face age-related musculoskeletal challenges such as joint stiffness, reduced flexibility, or arthritis pain, which can be exacerbated by everyday activities like gardening, walking, or even prolonged sitting. For example, a retiree who enjoys gardening may frequently experience knee pain from kneeling or lower back pain from bending over. Many retirees may not have regular access to physiotherapy services or prefer to manage their discomfort independently. A physiotherapy app could provide this group with low-impact exercises tailored to their needs, gentle stretching routines, and safe alternatives to strenuous movements. By incorporating reminders and progress tracking, the app could empower retirees to manage their discomfort proactively, maintaining their mobility and quality of life.

1.2.3 The Significance of the Product

- Empowers users with accessible, self-directed methods for managing and preventing physical discomfort.
- Provides guided tutorials, tailored exercise routines, and alternative recovery practices.
- Supports proactive musculoskeletal health across various lifestyles and professions.
    1. Athletes: Access to injury-prevention strategies and faster recovery support, helping to extend active years.
    2. Sedentary professionals (e.g., software developers): Offers ergonomic advice, tension-relief exercises, and reminders to move, reducing common issues like back and neck pain.
    3. Older adults and retirees: Access to safe, low-impact exercises for joint health and mobility, supporting active, independent lifestyles.
- Includes features like pain monitoring, real-time feedback, and movement education for comprehensive support.

- Serves as an on-demand physiotherapy assistant, reducing barriers to personalized care.
- Fills a gap in accessible, preventive physical care, fostering healthier, more comfortable lives.

### 1.2.4 The ‘Wow’ Factor

The 2D Model is a groundbreaking feature in the physiotherapy app, offering users an interactive way to pinpoint pain areas with an impressive level of anatomical detail. When users touch a specific area on the 2D body diagram, the app reveals a layered view of the underlying muscles, tissues, and joints, providing insights into the exact structures associated with their discomfort. This feature takes user experience to a new level, enabling a personalized exploration of pain points and potential sources of discomfort with just a few taps.

For example, if a user selects the lower back area, the app displays the muscle groups and tissues in that region. From there, the user can see information on potential strain points, typical pain causes, and even muscle imbalances that could be contributing to their discomfort. This interactive functionality empowers users to understand not only the location of their pain but also the possible reasons behind it—whether it’s muscle tension, joint stress, or inflammation.

By making complex anatomical information accessible in a visual and engaging way, the 2D Model brings a “wow” factor to self-care. It fosters a deeper understanding of the body, empowering users to make informed decisions about targeted exercises, stretching routines, and preventative measures. This feature transforms the app into a dynamic self-diagnostic tool, turning physiotherapy into an educational, intuitive, and highly interactive experience that builds user confidence in managing their physical health.

### 1.2.5 The Magnitude of the Problem

Physical pain and discomfort from musculoskeletal issues are becoming increasingly widespread due to sedentary lifestyles, high-stress work environments, and physically demanding occupations. According to the World Health Organization, lower back pain alone is one of the leading causes of disability worldwide, affecting over 540 million people at any given time. In the U.S., about 1 in 2 adults suffer from musculoskeletal conditions annually, which can hinder productivity, reduce quality of life, and even lead to long-term health complications if left untreated. Such conditions are not limited to manual laborers; office workers, software

developers, and athletes are equally susceptible due to prolonged sitting, repetitive movements, and intense physical demands.

For example, a software developer working extended hours at a desk might develop chronic neck pain and carpal tunnel syndrome, severely impacting their productivity and quality of life. Likewise, athletes who train rigorously often deal with repetitive strain injuries or ligament sprains, which, without prompt management, can escalate and lead to a decline in their performance. Retirees, too, face age-related musculoskeletal issues, such as arthritis and joint stiffness, which limit their mobility and independence.

Despite the growing need for effective management of these issues, many individuals lack easy access to professional physiotherapy, whether due to time constraints, location, or cost. This gap in accessible physical care highlights the necessity for an innovative solution that can provide immediate, personalized guidance. By addressing musculoskeletal pain and promoting proactive self-care, a physiotherapy app can significantly impact the well-being and productivity of diverse groups, creating a more accessible path to a healthier, pain-free life.

## 1.3 Problem Statement

Current physiotherapy services, while beneficial, lack the accessibility and personalization required to meet the diverse needs of patients suffering from musculoskeletal pain and injuries. This highlights the urgent need for a comprehensive platform that facilitates convenient and effective interaction between patients and physiotherapists.

## 1.4 Proposed Solution

The envisioned solution is an inventive virtual platform aimed at physiotherapy that eliminates existing barriers such as high costs, limited access to professionals, and challenges in maintaining consistency in treatment. At its core, the platform leverages technologies like Augmented Reality (AR) to guide users through exercises using 3D visual aids and real-time

feedback to ensure proper posture and alignment. While features like pain monitoring and progress tracking allow users to record their recovery journey, set objectives and make well-informed decisions regarding their rehabilitation, video consultations with physiotherapists provide remote access to professional counseling. These resources lessen reliance on in-person sessions by empowering users to take charge of their recovery.

The platform provides tailored injury remedies according to each user's needs, injury types and recovery status in order to guarantee individualized care. K-taping instructions, customized workout regimens, and first-aid procedures for efficient at-home recuperation are all available to users through the app. People may locate injuries, get detailed treatment recommendations and learn vital techniques through first-aid tutorials with the help of an easy-to-use 2D body model, people can identify the areas of injuries, get tailored treatment advice, and learn essential skills through first-aid lessons. A thorough collection of educational resources, gamification components and an AI chatbot for immediate physiotherapy advice are some extra features that keep users inspired and involved during their recovery process.

The platform prioritizes seamless integration and accessibility. With its electronic personal health record system, users can effortlessly store and share their medical history and recovery progress, ensuring that their care remains consistent. In addition, features like a nearby facilities locator and one-tap emergency contacts make it more convenient and safer for users.

In essence, this comprehensive and innovative solution aims to bridge the accessibility gap in physiotherapy, enabling athletes and everyday users to recover faster, avoid re-injury, and maintain long-term physical health. This encompassing approach aims to provide an efficient, affordable, and user-friendly physiotherapy experience.

## 1.5 Aim

The Goal is developing this physiotherapy app is to support and empower athletes and fitness enthusiasts in their journey to recover from certain injuries.Sports injuries,common injurie, can lead to long-term physical limitation if not treated properly.Our software provides a practical,personalized and customized approach to injury care and recovery,designed especially for people hurt while doing sports, doing exercises and doing their daily physical work.This app offers progress tracking, medical records of physio history of the user and scheduled workouts.

- 2D model : A 2D model of a human being to check and search the treatments by selecting the body part.
- EHR : Records of user's past physio injuries.
- Personalized workouts : Workout plans for each user regarding his EHR.

## 1.6 Project Scope

### 1.6.1 In-scope

Augmented Reality (AR) Guidance: Interactive 2D models that guide users through physiotherapy exercises with real-time feedback to ensure proper posture and technique.

Gamification: Points, badges, and progress milestones to incentivize users and improve adherence to physiotherapy routines.

Electronic Personal Health Record (EPHR): A secure digital record for users to store, view, and manage their medical and physiotherapy-related information.

Exercise Plan Customization: Ability for therapists to create and assign personalized exercise plans tailored to individual patient needs.

Remote Video Consultations: Integrated video calling for remote consultations between therapists and patients.

Progress Tracking: Monitoring tools to allow patients and therapists to track improvement over time and adjust treatment plans accordingly.

Notifications and Reminders: Alerts for scheduled exercises, medication reminders, and motivational tips to maintain user engagement.

### 1.6.2 Out-scope

Advanced Diagnostic Tools: This project will not include sophisticated diagnostic equipment or functionality to diagnose conditions, as the focus is on treatment rather than diagnosis.

Integration with Wearable Devices: The system will not integrate with wearable health devices (such as fitness trackers or smartwatches) due to technical and budget constraints.

## 1.7 Rich Picture Diagram

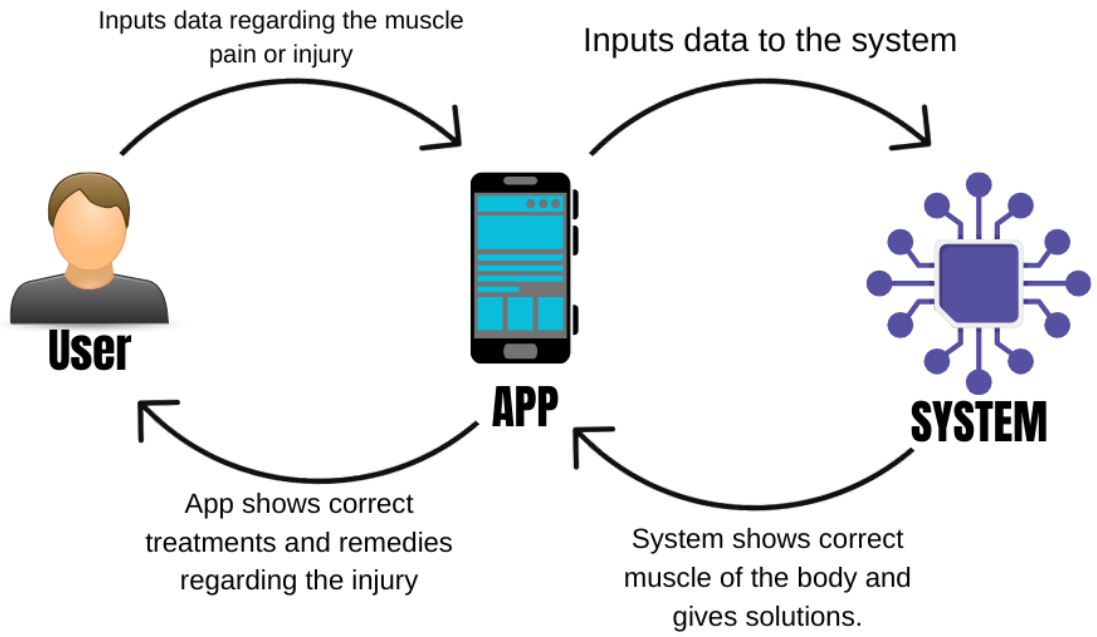



*Figure 1*

## 1.8 Resource requirements

Hardware Requirements

1. Devices

- User Access Device:The app will be accessed via mobile device. It's important to consider various screen sizes,resolutions, and processing capabilities to ensure a consistent user experience.
- Development Machines:Good quality computers with sufficient power, RAM and storage are required. These machines should be capable of handling tasks such as compiling code, running virtual environments, and testing app builds without lagging.

2. Server Hardware

- Application Servers:Reliable servers are necessary to host the application backend. These servers should have robust CPUs, adequate RAM, and sufficient storage to handle requests.
- Load balancing hardware:If the application expects high traffic, load balancers are needed to distribute user requests across multiple servers. This setup improves the app's responsiveness and prevents server overload.

3. Network Hardware

- Networking Infrastructure:Network hardware is required to support the development and operation of the application. This includes switches,routers and load balancers to keep the data flow and connectivity.

Software requirements

1. Operating Systems

- User platforms:The application should be compatible with both iOS and Android os to ensure that users can access the application on their devices.
- Development platforms:Developers need flexibility to work on different operating systems like Windows,MAC,Linux.

2. Development platforms and IDEs

- IDEs:VS Code,Android Studio,X code are popular IDEs for application development.These IDEs include simulators,debugging tools and various plugins.
- Cross platform frameworks:To develop the app for both Android and iOS with a single codebase frameworks like flutter,React Native and Xamarin are beneficial.

3. Emulators

- Android and iOS Emulators:Testing on real devices can be supplemented by using Android and iOS emulators,which simulate various devices, screens and operating system versions.This allow you to catch bugs and performance issues.

4. Version Control

- Git:Version control is essential for tracking changes,managing different versions of the application and to collaborate with team members.GitHub, GitLab provide a centralised repository where code can review,manage and merge.

## 1.9 Business model canvas

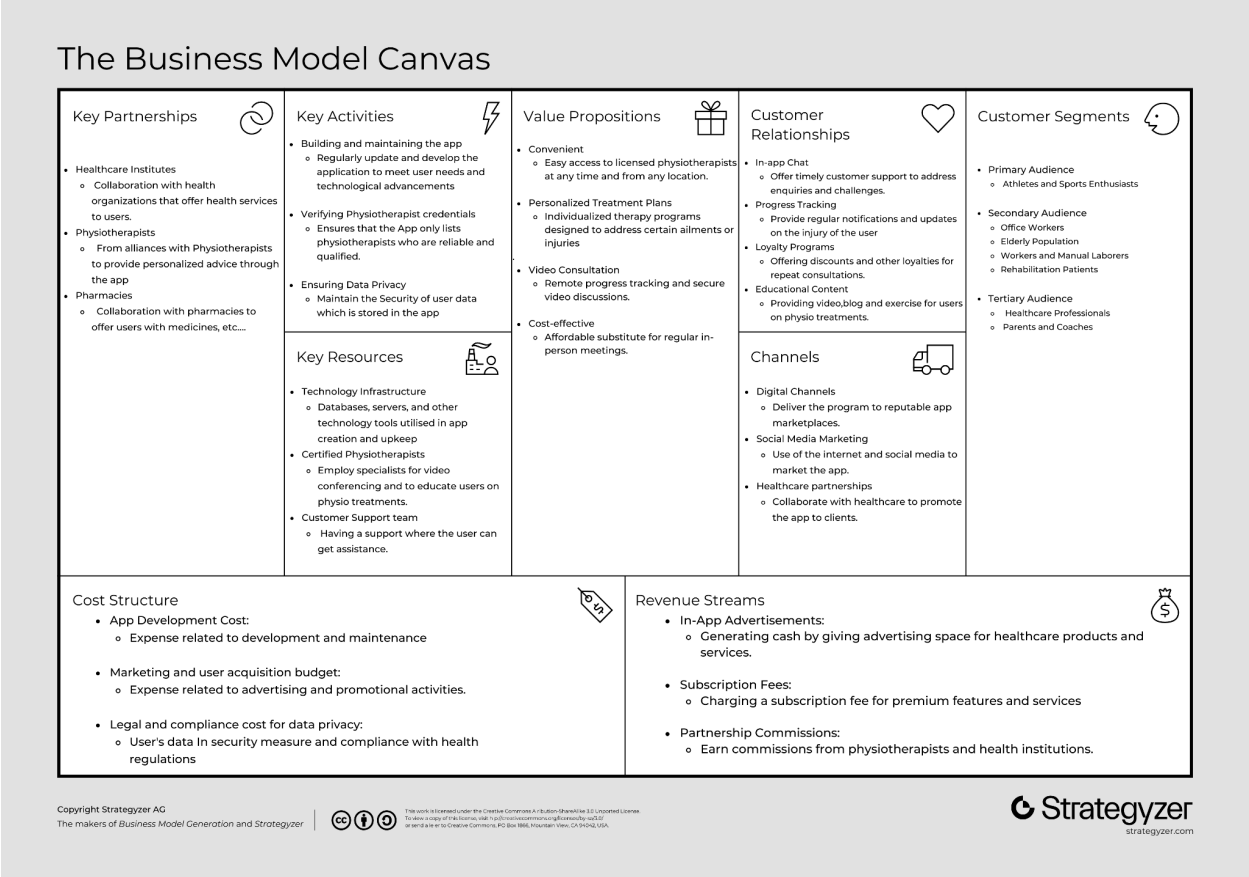



*Figure 2*

## 1.10 Chapter Summary

This chapter offered an overview of the project and explained the driving force for developing the PhysioConnect application. It stressed the increasing frequency of lifestyle-related physical

ailments, as well as the growing demand for accessible physiotherapy options. The project's objectives, scope, and significance were detailed, demonstrating its ability to address difficulties encountered by a variety of user groups, including athletes, employees, and the general public. Furthermore, the "wow factor" of innovative elements such as 2D models and AR-based consultations was highlighted, emphasizing the unique value that PhysioConnect provides. This chapter laid the groundwork for understanding the project's objectives and the issues it aims to address.

# Chapter 2: Existing work

## 2.1 Chapter Introduction

This chapter looks at existing physiotherapy apps such as Omada, Physitrack, SimpleTherapy, and PhysioConnect, emphasizing on their features, strengths, and limitations. While these apps offer useful features like video consultations and progress tracking, they lack creative features like interactive 2D models, augmented reality physiotherapy, daily recommendations, and one-tap emergency contacts. By benchmarking these systems, we identified critical gaps and possibilities to improve the user experience. This research guarantees that our software not only integrates existing platforms' best practices, but also provides new features, making it more broad and personalized to efficiently fulfill user needs.

## 2.2 Existing Work

There are already numerous apps on the market that are specifically designed for physiotherapy, each with its own strengths and drawbacks.An outline of some of our competitors and how our innovative PhysioConnect app outperforms them to offer a more complete and approachable solution for patients and physiotherapists can be shown below.

| Features | Omada | Physitrack | SimpleTherapy | PhysioConnnect (Our Product) |
|---|---|---|---|---|
| Video Consultation | ✅ | ✅ | ✅ | ✅ |
| Progress Tracking | ✅ | ✅ | ✅ | ✅ |
| Educational Resources | ✅ | ✅ | ✅ | ✅ |
| Customizable Exercise Plans | ✅ | ✅ | ✅ | ✅ |
| Pain Monitoring | ✅ | ✅ | ✅ | ✅ |
| In-app Messaging | ✅ | ✅ | ❌ | ✅ |
| Gamification | ✅ | ✅ | ❌ | ✅ |
| EPHR(Electronic Personal Health Record) | ✅ | ✅ | ❌ | ✅ |
| AI Chatbot for Physiotherapy Advice | ✅ | ❌ | ❌ | ✅ |
| Nearby Facilities Locator | ✅ | ❌ | ❌ | ✅ |
| 2D Model Tutorial | ❌ | ❌ | ❌ | ✅ |
| Daily Tip Generator | ❌ | ❌ | ❌ | ✅ |
| One-Tap Emergency Contact | ❌ | ❌ | ❌ | ✅ |
| AR-based Physiotherapy Sessions | ❌ | ❌ | ❌ | ✅ |

*Table 2*

2.2.1 Review on benchmarking

To get a detailed analysis of how PhysioConnect differentiates from the other competitors in the same market, the benchmarking process helped us to highlight how we will fill the gap using our unique solution.

#### 2.2.1.1 Common Features

Most of the competitors provide the most essential features,

- **Video Consultation:** A cornerstone for telehealth solutions, enabling remote communication with physiotherapists.
- **Progress Tracking:** Tools to monitor a user's rehabilitation or fitness journey.
- **Educational Resources:** Access to videos and articles explaining physiotherapy practices.
- **Customizable Exercise Plans:** Personalized exercise regimens tailored to user conditions.

#### 2.2.1.2 USP of our App

What sets our product apart are the innovative features aimed at enhancing usability, engagement, and accessibility:

**Pain Monitoring:** Unlike SimpleTherapy and PhysioConnect, our app provides tools to track specific pain areas and progression over time, offering better symptom management.

**Gamification**: Encourages user engagement through rewards and challenges, an element missing in some competitors like PhysioConnect

**EPHR (Electronic Personal Health Record):** Ensures that users have a centralized and secure repository of their medical and physiotherapy data.

**AI Chatbot for Physiotherapy Advice:** Offers instant responses for user queries, an advantage over Physitrack and SimpleTherapy.

**Nearby Facilities Locator:** A Google Maps integration enabling users to find clinic, ayurvedic medicine and home-maid remedies and pharmacies with physiotherapy supplies or remedies.

**2D Model Tutorial:** A groundbreaking feature allowing users to interactively identify and understand pain points and get specified advice with the most accurate muscle/tissue injured, absent in all compared solutions.

**Daily Tip Generator:** Encourages micro-interventions by providing simple, actionable advice to improve health daily.

**One-Tap Emergency Contact:** Prioritizes user safety by enabling immediate connection to pre-configured emergency contacts or any nearby contact.

**AR-based Physiotherapy Sessions:** An advanced feature revolutionizing virtual therapy by allowing users to interact with physiotherapy routines in augmented reality.

2.2.2 Technology/Approaches/Algorithms Review

**Technologies**

**Augmented Reality (AR):** Utilized for guiding users through physiotherapy exercises with real-time posture correction and visual aids. AR enhances user interaction and promotes better adherence to recovery routines.

**2D Human Body Model:** Interactive visualization for pinpointing pain areas and providing tailored recommendations based on muscle groups and injury types.

**Artificial Intelligence (AI)**: Powers the chatbot for instant physiotherapy advice and tailors user experiences by analyzing past data, preferences, and recovery trends.

**Electronic Health Records (EHR):** Centralized storage of user health history, ensuring easy access and better-informed recovery plans.

**Cloud Storage:** Ensures secure and scalable data management for user records, app content, and progress tracking.

**Approaches:**

**User-Centered Design (UCD):** Focuses on creating a user-friendly interface by involving end-users in the design and development process.

**Lean UX:** Combines fast iterations, hypothesis testing, and collaborative workflows to optimize the app's design and functionalities.

**Agile Methodology:** Ensures a flexible, iterative development process that accommodates feedback and evolving requirements.

**Algorithms:**

**Posture Correction Algorithm:** Uses AR and motion sensors to assess user movements and provide real-time feedback.

**Recommendation System:** Suggests exercises, stretches, and remedies based on user data, past injuries, and recovery progress.

**Progress Tracking Algorithm:** Monitors user activities and recovery metrics, providing insights and visualizing improvements over time.

**Emergency Contact Algorithm:** Ensures one-tap connection to predefined contacts or facilities in case of urgent assistance needs.

**Gamification Algorithm:** Keeps users engaged by rewarding progress and encouraging adherence to physiotherapy routines through badges or points.

#### 2.2.3 Testing and evaluation metrics

Our main emphasis is in order to provide sufficient care and a crash-proof application are;

- User Retention - Aids in understanding user loyalty and happiness, provides indication of how user proportions are regarding the usage of the application.
- Recovery Progress - Tracks users reported gains in mobility or discomfort, showing how well the software will aid in recovery and rehabilitation.
- Feature Acceptance - Analysis of the more used features and which are less used and have potential to improve. Constant feedback regarding every feature allowing more utilised features more improved facilities in the software.

## 2.3 Tools and Implementation plan

### 2.3.1 Technology stack and Development Approach

#### 2.3.1.1 Design Methodology

Integrating Lean UX and UCD within Agile methodologies like Scrum ensures a strong, user-focused development process. By combining the strengths of both approaches, speed and user satisfaction can be balanced, while leading to more innovative and effective solutions.

#### 2.3.1.2 Development Methodology

Agile development with a Scrum framework is an excellent option since it allows for flexibility and adaptability to changing requirements. In case of a health-focused app, where user feedback and iterative improvements are essential to succeed, this strategy guarantees frequent feedback.

#### 2.3.2 Core Technologies and Features

- Front End: Flutter for highly responsive, interactive and cross-platform UI development.

- Back End: Node.js and Express.js for efficient server-side operations, data handling and API integration.
- Database: MangoDB for scalable and efficient data management.
- Security: To ensure the confidentiality of sensitive user data, the app employs end-to-end encryption, OAuth 2.0 authentication, HIPAA/GDPR compliance, role-based access control, biometric authentication, and real-time monitoring with tools like AWS Shield to prevent threats.
- UI /UX: Figma for crafting, engaging visuals and user-friendly design elements.
- Testing and Quality Assurance: Tools like Firebase, and continuous delivery (CI/CD) are used for performance optimization. Accessibility testing ensures compliance with WCAG standards, while user feedback and usability testing with Figma prototypes refine functionality and design.

2.3.3 Implementation Plan

1. **Prototype Development**

   Prototype development is a critical phase in the implementation plan of any project. It involves creating a preliminary model of the product, validating ideas , gathering feedback, and refine the design. Here is a breakdown of how to implement prototype development effectively.

2. **Testify and Refinement**

   The testify and refinement phase is where the prototype or product undergoes rigorous testing to ensure it meets the desired standards, followed by iterative improvements based on feedback . This step is essential for delivering a high-quality product that aligns with user needs and project goals.

## 2.4 Chapter Summary

The emphasis of this chapter is still on defining the crucial elements and tactics required for a more successfully integrated software implementation. System's introduction, examines current digital physiotherapy technologies and solutions, assesses important testing and performance indicators, and lists the resources and thorough implementation strategy.
A solid and user-centric solution is made possible by this methodical approach, which guarantees that every aspect of this PhysioConnect application is guided by in-depth research and thorough execution strategies.

# Chapter 3: Methodology

## 3.1 Chapter Overview

The research approach used for this project is thoroughly analysed in this chapter, which also includes information on the development design and project management techniques that were chosen. It includes how these methods reduce risks, maximise resource use, optimise the execution process, and enhance the project's overall success.

## 3.2 Development Methodology

Agile and Scrum are ideal for the development of the PhysioConnect since they emphasize iterative progress and user-centric features. Agile provides a structure for continuous development, allowing the team to release the software in smaller, more manageable chunks. This method is vital for a project with a variety of features, such as AR-based consultations, 2D training, and pain monitoring, because it allows the team to prioritize critical capabilities while refining the app based on continuing input. Scrum complements Agile by offering defined sprint cycles in which specified tasks, such as developing the daily tip generator or the emergency

contact system, are performed within set timescales. This ensures that features are delivered on schedule and to high quality standards.

## 3.3 Design methodology

For this project, we adopt a User-Centred Design (UCD) methodology combined with Lean UX principles. This hybrid approach is chosen to ensure the final product meets the users' needs while maintaining an efficient and iterative design process.

UCD focuses on involving the users at every stage of the design process, ensuring the app is intuitive and meets their expectations. This is particularly relevant for a physiotherapy platform, as it requires usability and accessibility to cater to diverse user needs, including injured individuals and medical professionals.

The UCD process involves understanding the target audience through interviews, surveys, and personas to identify pain points and desired features. Wireframes and prototypes are created to test usability, with feedback integrated iteratively to refine the design and ensure it meets user needs effectively.

Lean UX aligns with Agile development and is perfect for a tech-driven project with fast iteration cycles. It emphasizes collaboration, quick prototyping, and minimizing waste, ensuring rapid delivery of features without compromising on quality. (Mirafshar, 2023)

Lean UX emphasizes cross-functional collaboration among designers, developers, and stakeholders to align goals. It uses hypothesis-driven design to validate features through user testing, focuses on delivering essential features with Minimal Viable Design (MVD), and follows an iterative process with frequent updates based on real-world feedback.

## 3.4 Project Management Methodology

For the PhysioConnect project the Agile-Kanban technique was selected because of its adaptability, emphasis on continuous delivery, and visual workflow management. Agile-Kanban

is perfect for a dynamic project like this where iterative development and changing user needs are crucial since it blends the concept of agility with the organised task visualisation of Kanban.

Agile-Kanban;s flexibility is one of its strongest draws. With stages like “To Do”,”In Progress” and “Completed” tasks are maintained on a Kanban technique, giving a clear picture of the workflow. Team members and stakeholders may readily monitor progress. Furthermore, the team may adapt to requirements changes without interfering with workflow since there are no set sprint cycles, which permits more flexible prioritisation.

All things considered, PhysioConnect’ iterative and user-centric design fits in nicely with Agile-Kanban. Its adaptability facilitates quick feature updates and the integration of real-time input, and its visual organisation encourages efficiency and teamwork.

## 3.5 Team Work Breakdown Structure (WBS)

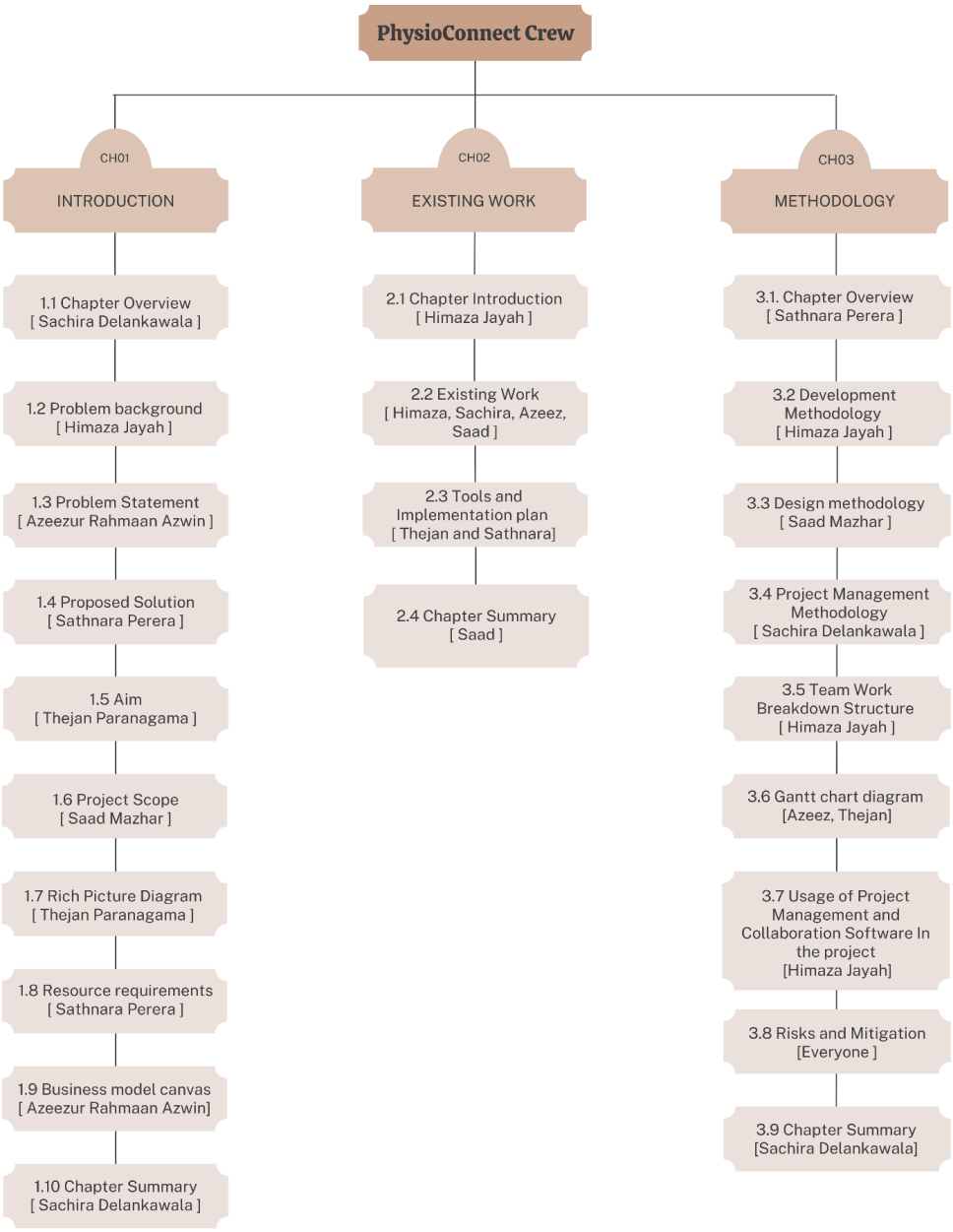
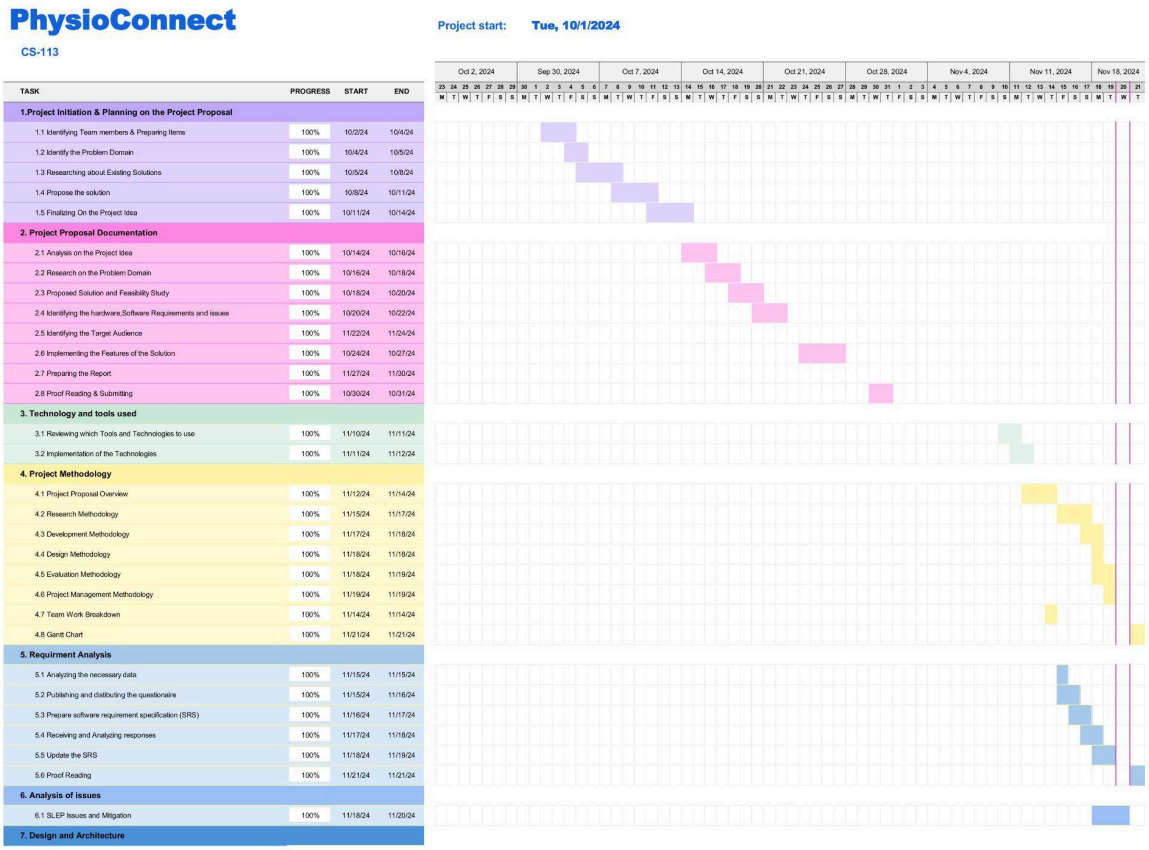



*Figure 3*

PhysioConnect_CS_113

## 3.6 Gantt chart diagram

Pragathi Paranagama Rahman Azwin



**PhysioConnect**

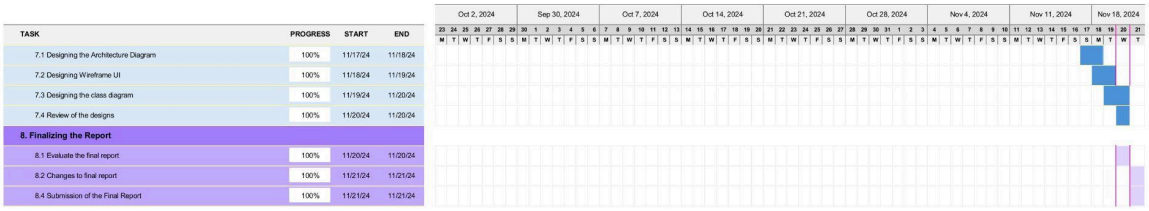
CS-113

Project start: Tue, 10/1/2024

| TASK | PROGRESS | START | END |
|---|---|---|---|
| **1.Project Initiation & Planning on the Project Proposal** | | | |
| 1.1 Identifying Team members & Preparing Items | 100% | 10/2/24 | 10/4/24 |
| 1.2 Identify the Problem Domain | 100% | 10/4/24 | 10/5/24 |
| 1.3 Researching about Existing Solutions | 100% | 10/5/24 | 10/8/24 |
| 1.4 Propose the solution | 100% | 10/8/24 | 10/11/24 |
| 1.5 Finalizing On the Project Idea | 100% | 10/11/24 | 10/14/24 |
| **2. Project Proposal Documentation** | | | |
| 2.1 Analysis on the Project Idea | 100% | 10/14/24 | 10/16/24 |
| 2.2 Research on the Problem Domain | 100% | 10/16/24 | 10/18/24 |
| 2.3 Proposed Solution and Feasibility Study | 100% | 10/18/24 | 10/20/24 |
| 2.4 Identifying the hardware,Software Requirements and issues | 100% | 10/20/24 | 10/22/24 |
| 2.5 Identifying the Target Audience | 100% | 11/22/24 | 11/24/24 |
| 2.6 Implementing the Features of the Solution | 100% | 10/24/24 | 10/27/24 |
| 2.7 Preparing the Report | 100% | 11/27/24 | 11/30/24 |
| 2.8 Proof Reading & Submitting | 100% | 10/30/24 | 10/31/24 |
| **3. Technology and tools used** | | | |
| 3.1 Reviewing which Tools and Technologies to use | 100% | 11/10/24 | 11/11/24 |
| 3.2 Implementation of the Technologies | 100% | 11/11/24 | 11/12/24 |
| **4. Project Methodology** | | | |
| 4.1 Project Proposal Overview | 100% | 11/12/24 | 11/14/24 |
| 4.2 Research Methodology | 100% | 11/15/24 | 11/17/24 |
| 4.3 Development Methodology | 100% | 11/17/24 | 11/18/24 |
| 4.4 Design Methodology | 100% | 11/18/24 | 11/18/24 |
| 4.5 Evaluation Methodology | 100% | 11/18/24 | 11/19/24 |
| 4.6 Project Management Methodology | 100% | 11/19/24 | 11/19/24 |
| 4.7 Team Work Breakdown | 100% | 11/14/24 | 11/14/24 |
| 4.8 Gantt Chart | 100% | 11/21/24 | 11/21/24 |
| **5. Requirment Analysis** | | | |
| 5.1 Analyzing the necessary data | 100% | 11/15/24 | 11/15/24 |
| 5.2 Publishing and distibuting the questionaire | 100% | 11/15/24 | 11/16/24 |
| 5.3 Prepare software requirement specification (SRS) | 100% | 11/16/24 | 11/17/24 |
| 5.4 Receiving and Analyzing responses | 100% | 11/17/24 | 11/18/24 |
| 5.5 Update the SRS | 100% | 11/18/24 | 11/19/24 |
| 5.6 Proof Reading | 100% | 11/21/24 | 11/21/24 |
| **6. Analysis of issues** | | | |
| 6.1 SLEP Issues and Mitigation | 100% | 11/18/24 | 11/20/24 |
| **7. Design and Architecture** | | | |

*Figure 4*

PhysioConnect_CS_113



| TASK | PROGRESS | START | END |
|---|---|---|---|
| 7.1 Designing the Architecture Diagram | 100% | 11/17/24 | 11/18/24 |
| 7.2 Designing Wireframe UI | 100% | 11/18/24 | 11/19/24 |
| 7.3 Designing the class diagram | 100% | 11/19/24 | 11/20/24 |
| 7.4 Review of the designs | 100% | 11/20/24 | 11/20/24 |
| **8. Finalizing the Report** | | | |
| 8.1 Evaluate the final report | 100% | 11/20/24 | 11/20/24 |
| 8.2 Changes to final report | 100% | 11/21/24 | 11/21/24 |
| 8.4 Submission of the Final Report | 100% | 11/21/24 | 11/21/24 |



*Figure 5*

## 3.7 Usage of Project Management and Collaboration Software In the project

### 3.7.1 Google Sheets for Task Allocation

| Chapter 1: Introduction | | | | | | |
|---|---|---|---|---|---|---|
| **TASK** | **Sachira** | **Himaza** | **Azeez** | **Sathnara** | **Thejan** | **Saad** |
| **1.1 Chapter Overview** | | | | | | |
| This is a small introduction to your chapter. | | | | | | |
| **1.2 Problem background** | | | | | | |
| *convince the reader why this project must be carried out,* | | | | | | |
| **1.3 Problem Statement** | | | | | | |
| *Problem, in a nutshell just one sentence* | | | | | | |
| **1.4 Proposed Solution** | | | | | | |
| *Explanation of the proposed solution* | | | | | | |
| **1.5 Aim** | | | | | | |
| *One sentence in italics* | | | | | | |
| **1.6 Project Scope** | | | | | | |
| *Include the things you will do & won't do as part of your final* | | | | | | |
| **1.7 Rich Picture Diagram** | | | | | | |
| *It should be very simple and easy to understand.* | | | | | | |
| **1.8 Resource requirements** | | | | | | |
| *Resources needed to work on the minimum viable product* | | | | | | |
| **1.9 Business model canvas** | | | chickenpox but done | | | |
| *download - https://assets-global.website-files.com/64830736e* | | | | | | |
| **1.10 Chapter Summary** | | | | | | |
| *This is a small summary of your chapter.* | | | | | | |

*Figure 6*

| Chapter 2: Existing work - *optional subtopics* | | | | | | |
|---|---|---|---|---|---|---|
| **TASK** | **Himaza** | **Sachira** | **Sathnara** | **Azeez** | **Thejan** | **Saad** |
| 2.1 Chapter Introduction | | | | | | |
| | | | | | | |
| 2.2 Existing Work | | | | | | |
| | | | | | | |
| 2.3 Tools and Implementation plan | | | | | | |
| | | | | | | |
| 2.4 Chapter Summary | | | | | | |

*Figure 7*

| Chapter 3: Methodology | | | | | | |
|---|---|---|---|---|---|---|
| **TASK** | **Himaza** | **Sachira** | **Sathnara** | **Azeez** | **Thejan** | **Saad** |
| 3.1. Chapter Overview | | | | | | |
| | | | | | | |
| 3.2 Development Methodology | | | | | | |
| | | | | | | |
| 3.3 Design methodology | | | | | | |
| | | | | | | |
| 3.4 Project Management Methodology | | | | | | |
| | | | | | | |
| 3.5 Team Work Breakdown Structure (WBS) | | | | | | |
| | | | | | | |
| 3.6 Gantt chart diagram | | | | | | |
| | | | | | | |
| 3.7 Usage of Project Management and Collaboration Softw | | | | | | |
| | | | | | | |
| 3.8 Risks and Mitigation | | | | | | |
| | | | | | | |
| 3.9 Chapter Summary | | | | | | |

*Figure 8*

### 3.7.3 Slack for important updates and announcements

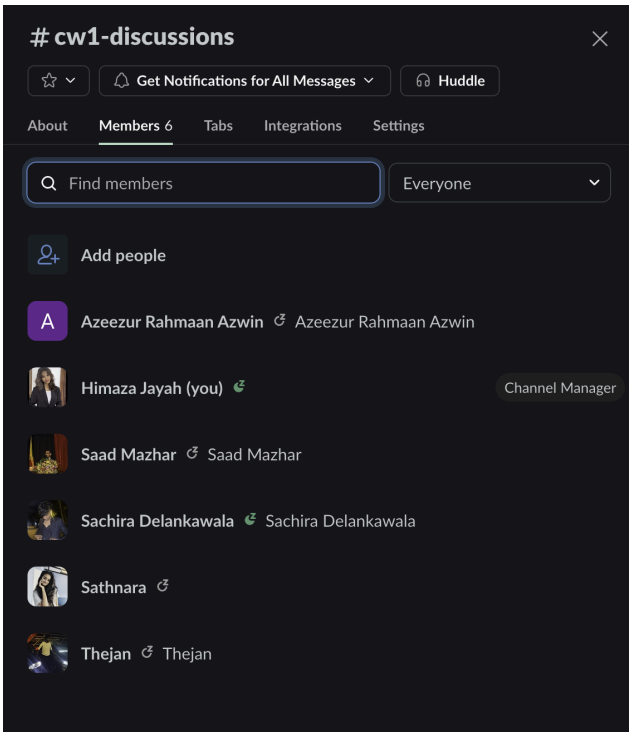



*Figure 9*

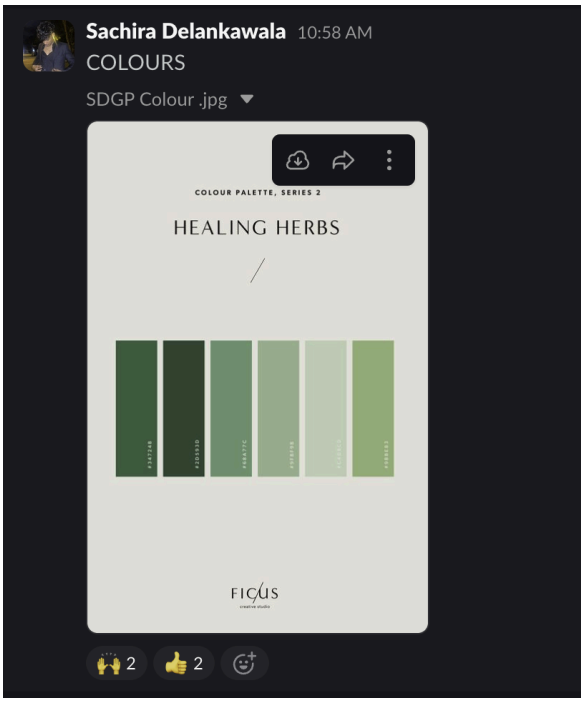



*Figure 10*

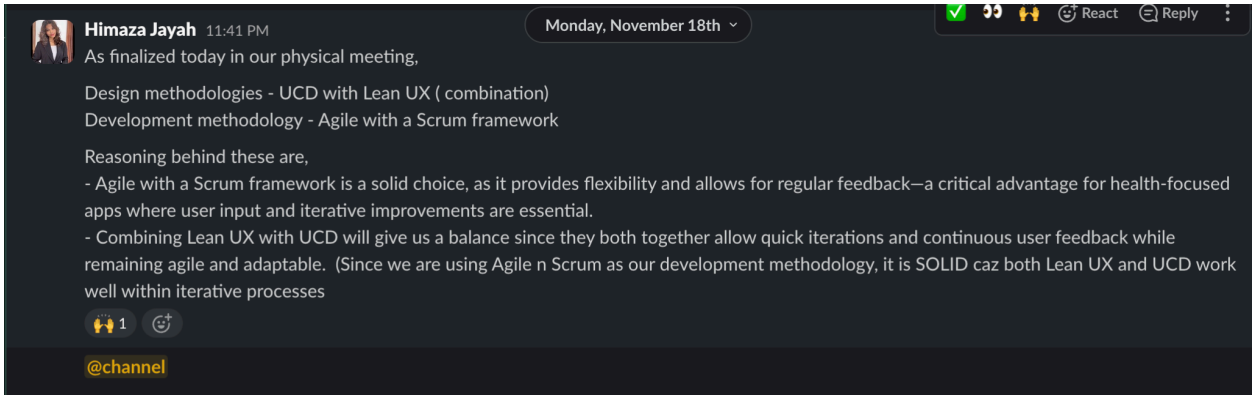



*Figure 11*




*Figure 12*




*Figure 13*

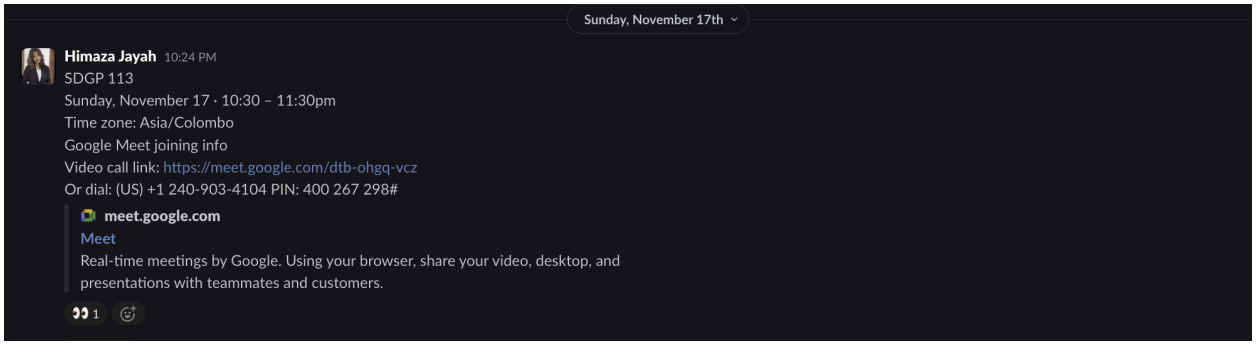



*Figure 14*

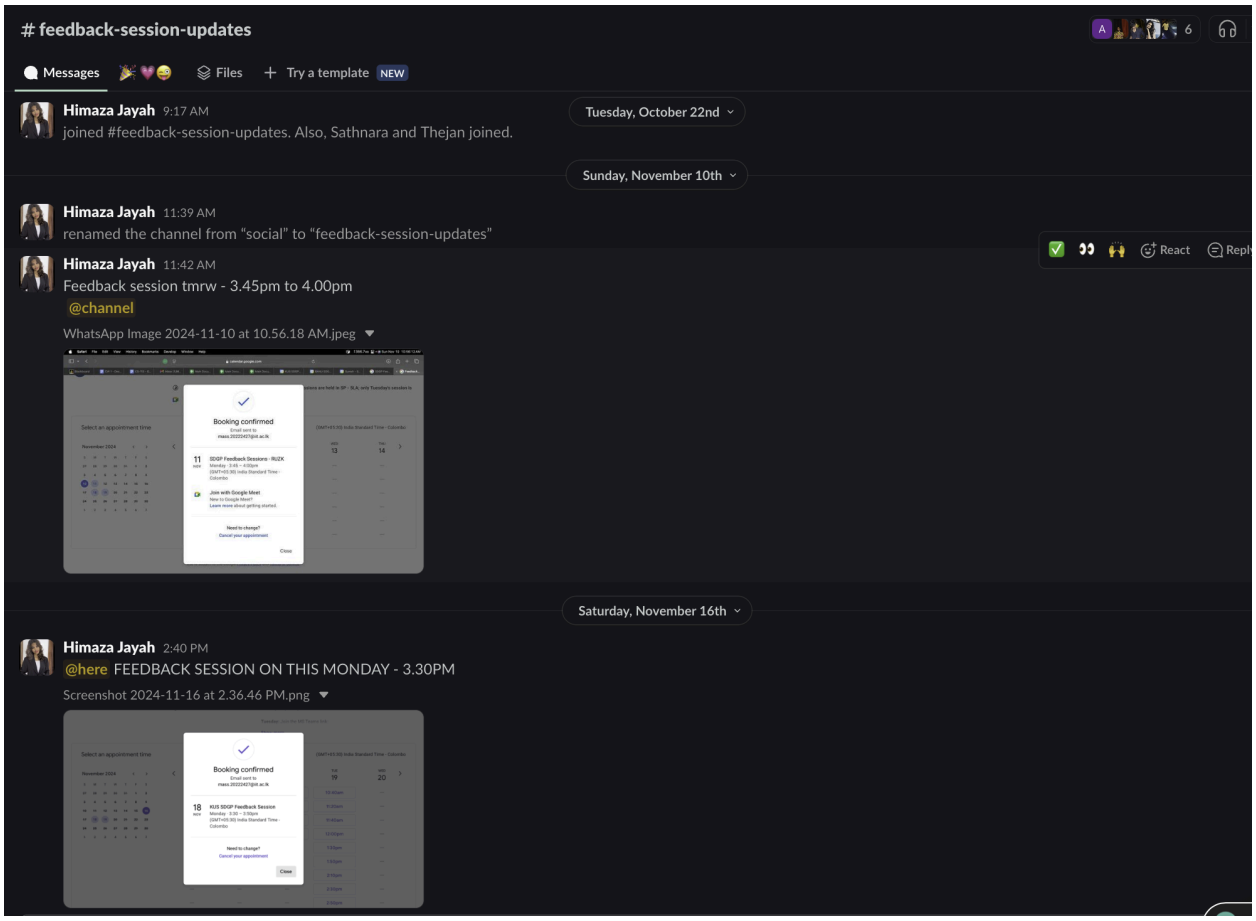



*Figure 15*

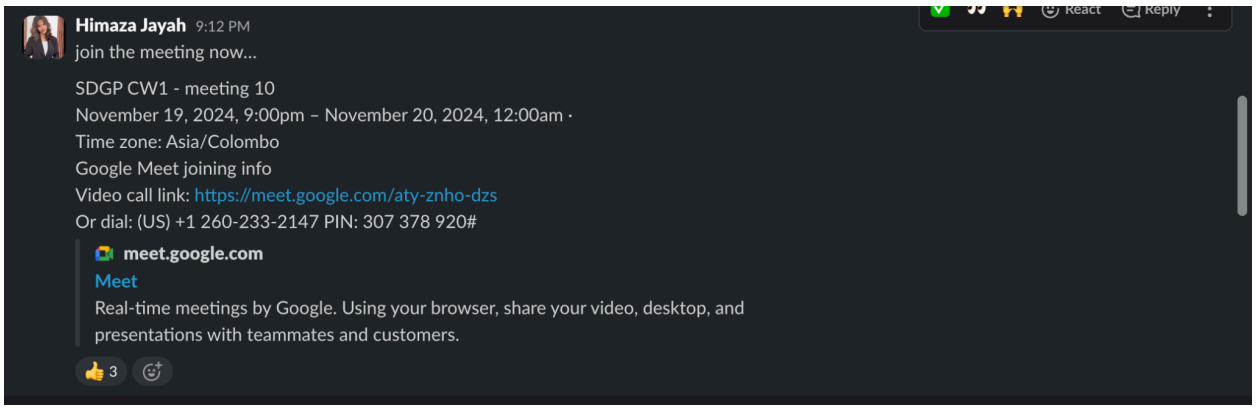



*Figure 16*

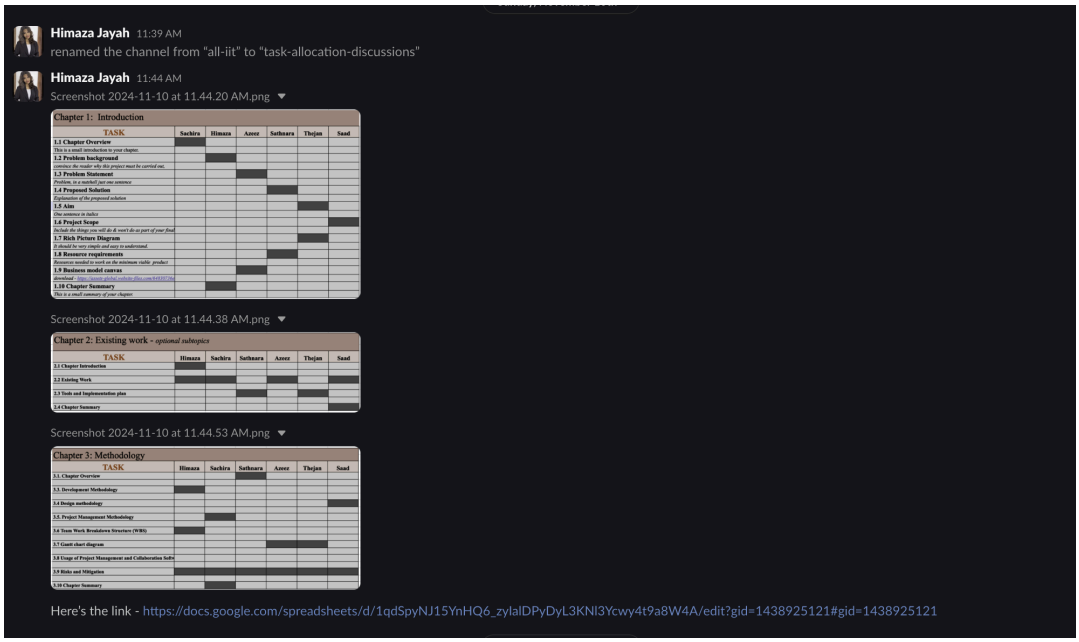



*Figure 17*

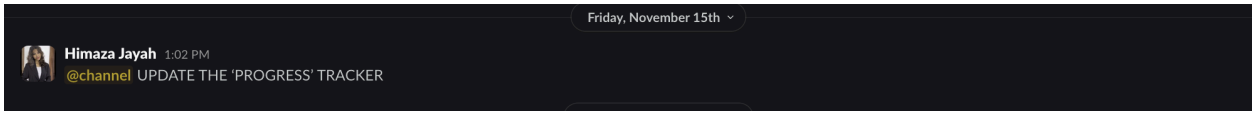
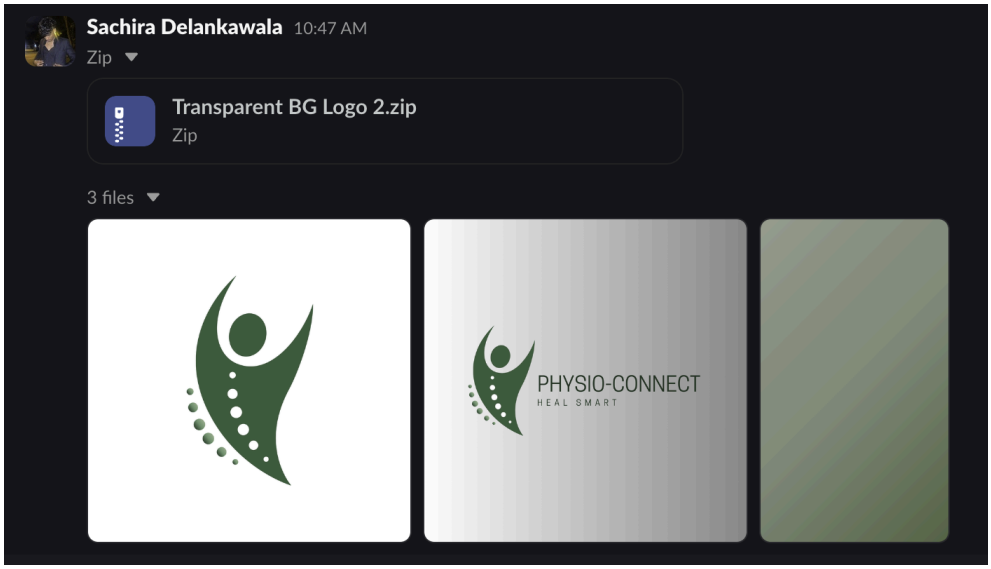



*Figure 18*




PhysioConnect_CS_113

*Figure 19*

### 3.7.3 Googlemeets for important discussions and finalizations

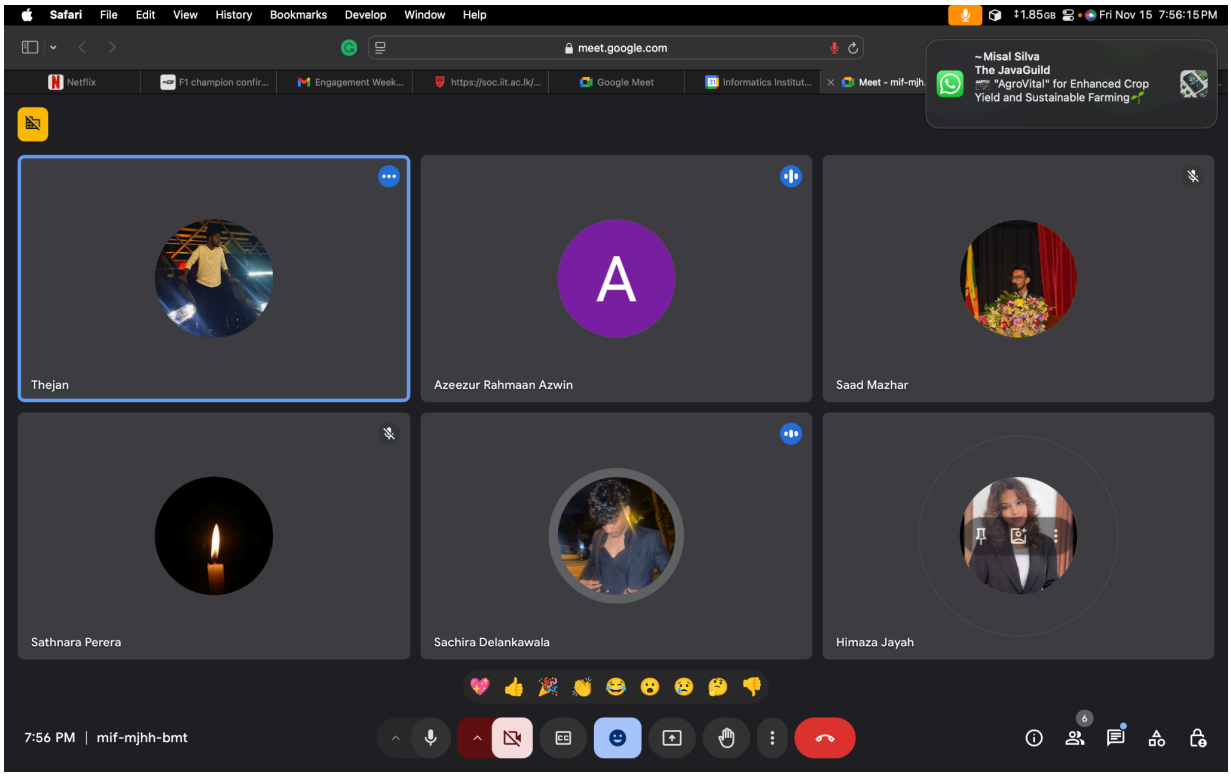



*Figure 20*

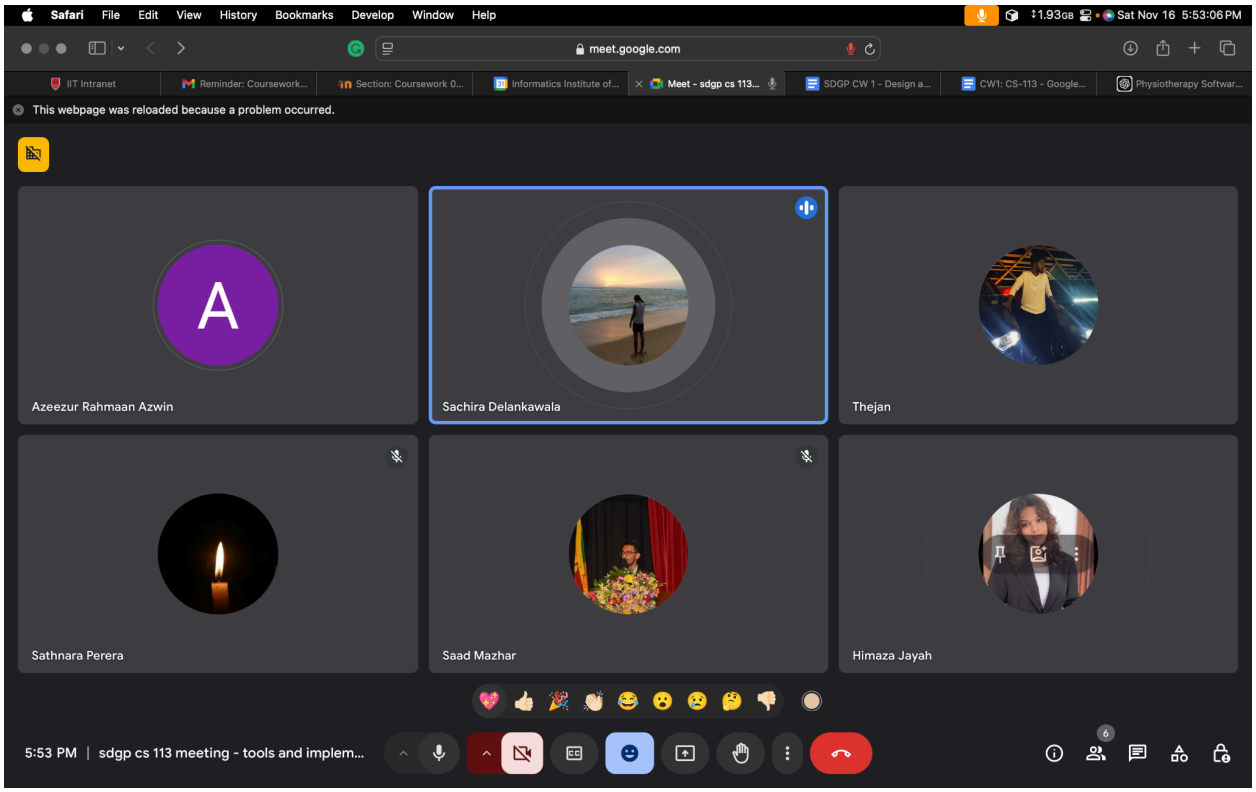



*Figure 21*

## 3.8 Risks and Mitigation

| **Risk Item** | **Severity** | **Frequency** | **Mitigation Plan** |
|---|---|---|---|
| Resource Limitations | **5** | **5** | Adopt Agile methodology to prioritize critical features, allocate resources effectively, and seek expert consultation when necessary. |
| Data breaches, privacy and security risks | | | - Implement strong data encryption<br>- Regular Security audits.<br>- Adherer to GDPR and HIPAA compliance standards |

| | | | |
|---|---|---|---|
| Miscommunication among Team Members | | | Schedule regular team meetings.<br>Use project management tools like Google Sheets or Slack for clear communication.<br>Maintain detailed documentation to ensure everyone is aligned. |
| Technology Limitations | | | Optimise features for performance, provide fallback options for lower-end devices, and test the app across diverse hardware configurations. |
| Conflicts over decisions regarding design and feature prioritisation | | | Use a democratic decision-making process where team members vote on contentious issues, and involve a mediator or project lead to resolve disputes objectively. |
| | | | |

*Table 3*

## 3.9 Chapter Summary

The PhysioConnect project methodology, with an emphasis on development, design, and project management techniques, is described in this chapter. Because of its user-centric, iterative methodology, which facilitates rapid feature delivery and ongoing development, the Agile-Scrum framework was chosen. When paired with Lean UX principles, the User-centred Design (UCD) methodology guarantees that the platform is user-friendly and efficiently serves a variety of user demands. The Agile-Kanban approach was selected for project management because it provides continuous delivery, flexibility, and visual task management. The chapter also emphasises the use of project management tools, hazards, and mitigation techniques to guarantee the platform is delivered successfully and with the least amount of disruption.

# 12. References

*Musculoskeletal (MSK) program | OMadA Health*. (n.d.). https://www.omadahealth.com/musculoskeletal

Rantala, A. (n.d.). *Physitrack® - The world leader in remote patient engagement and Telehealth*. https://www.physitrack.com/

*SimpleTherapy: Personalized Virtual & In-Person MSK & Behavioral Health Care. | SimpleTherapy*. (n.d.). https://www.simpletherapy.com/en/

UX, L. (2023). Lean UX Process: Integrating UX into Agile Development. [online] YouTube. Available at: https://youtu.be/Y61MZd3v2Yc?si=3ww0C7n8S8V48I_G [Accessed 20 Nov. 2024].

*UML activity diagram: Activities and connections* (2024) *SparxSystems Europe*. Available at: https://www.sparxsystems.eu/languages/uml/diagrams/activitydiagram/ (Accessed: 20 November 2024).

Stewart, A., Hoyt, T. and Armstrong, C.M. (2018) *(PDF) mobile applications for client use: Ethical and legal considerations*, *Mobile Applications for Client Use: Ethical and Legal Considerations*. Available at: https://www.researchgate.net/publication/329821254_Mobile_Applications_for_Client_Use_Ethical_and_Legal_Considerations (Accessed: 16 November 2024).

Peek, Sean. "What Is Agile Scrum Methodology?" Business News Daily, 2023, www.businessnewsdaily.com/4987-what-is-agile-scrum-methodology.html [Accessed on 15th November 2024]

# 13. Appendix